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Singapore Mass Rapid Transit System: design

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The development of the system concepts is described. The principles involved in the layout of the stations are detailed and examples of the various types that evolved are given. The design for fire and emergency is highlighted. The longitudinal profile is described in relation to the tunnels, the at-grade, and the elevated sections with particular reference to energy conservation. The influence of advances in rolling-stock technology are mentioned. Alignment parameters are evaluated. The permanent way options that were considered are outlined, the attention given to noise suppression noted and the final choice of three specific types is described. The requirements dictating the form and location of other structures are mentioned. The effect of geotechnical considerations on design is described. The effect of the contract forms—design and construct, consultants and in-house engineers' design is noted and the design organization derived from these considerations to conform with Singapore legislation is given. The principal design features of the system follow with particular attention to tunnel linings, civil defence underground station structures, viaduct types and the generic types of elevated stations developed for the four main above ground sections. The main depot and two subsidiary stabling sidings posed individual problems and the design solutions for each are given. The size and speed of the project has afforded an opportunity to compare various design solutions to many aspects of MRT design and some comparisons are made and preliminary conclusions drawn.

Concepts

The brief presented to the Singapore Mass Rapid Transit (MRT) Corporation by government for the design of the mass transit railway was unequivocal. The system was to be the most modern available compatible with the use of components that had been tried and tested in service under conditions similar to those anticipated for the Singapore system. The system had to incorporate the latest technology but it was not considered realistic to devote resources to research and development in mass transportation. The parameters used were therefore those developed by other efficient systems.

2. Consultants from North America, the UK, continental Europe and Asia

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were involved at various stages and the MRT Corporation recruited specialist staff overseas largely from the UK and Hong Kong (both British and Chinese). However, the general direction of the project and its senior management were entirely local.

3. The design life was stipulated as 75-100 years. Coupled with this was a requirement that the design should reduce maintenance to an absolute minimum.

Traffic data

4. Traffic studies carried out in phases I and II of the Mass Rapid Transit (MRT) project and subsequently updated provided figures for each interstation section for the morning and evening 90 min peak periods and for the 13 h off-peak periods, for service dates up to beyond 2000. To define the maximum passenger flows to be designed for, a factor of 1·4, derived from data from other comparable systems, was used to calculate the 15 min peak traffic flow rate; other factors varying from 0·4 to 1·35 were used to calculate various off-peak periods and weekend usage.

Train formation

5. It was considered that a maximum train frequency of 32 per h was comfortably within the capabilities of modern train signalling systems. The maximum train load was set at 2/3 of the crush capacity of trains—some allowance being made for the generally slighter stature of Asians and the lighter clothing of a tropical city—and a commercial speed (i.e. allowing for station stops, and so on) of 40 km/h was adjudged necessary to make the system attractive. The maximum speed was set at 80 km/h and the nominal code speeds for automatic train operation were 64, 48 and 20 km/h

6. Based on the traffic studies a train with a crush capacity of 1800 was developed as suitable to meet all the long-term forecasts. Increases in system capacity would be obtained by increasing train frequencies as necessary up to the 32 per h maximum. Analysis of rolling stock worldwide confirmed the trend towards longer cars and a length over buffers of 23 m was adopted. Similarly, train widths of 3 m or slightly greater appeared the most widely adopted and a dimension of 3·2 m was selected. From these dimensions a six car train formation evolved to meet the required capacity (Fig. 1).

Traction power supply

7. Phase III of the MRT study recommended an overhead 1500 V DC traction power supply and the technical superiority of the catenary system was clearly recognized. However, Singapore's concern to provide an aesthetic environment, particularly in its public works, led eventually to a decision to adopt a 750 V DC third rail traction supply system to avoid a proliferation of support standards and wires.

Structure gauge

8. From the notional load gauge the kinematic envelope was developed by consideration of the rolling stock suspension movements, wear and tear, vehicle yaw and pitch, and track tolerances. A structure gauge with a side clearance of 100 mm and a minimum overhead clearance of 70 mm to the kinematic envelope was adopted (Fig. 2). It was considered that detraining must always be under the train staff's control and as this can only be achieved by exiting through the train end

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Fig. 1. Train

doors, a walkway was not provided. The need to provide adequate room for floating slab track in tunnels in sensitive areas to minimize nuisance from vibration was accepted early and the depth needed for its construction led to the adoption of a 5·2 m i.d. tunnel.

Energy conservation

9. In a state with no natural resources energy costs are principally dictated by world prices. Energy conservation thus becomes even more important and its benefits should extend throughout the working life of the railway. Two aspects of the system were particularly significant, namely alignment and the use of platform screen doors.

10. *Alignment.* The emphasis on energy conservation led naturally to the benefits to be derived from so-called hump and saw-tooth profiles being examined; the introduction on modern rolling-stock of regenerative braking reduces but does not eliminate the savings to be obtained. On at-grade and viaduct sections, however, the disadvantages of having to lift stations 2–3 m above the general track level were felt to outweigh the energy and run time advantages. In cut-and-cover tunnels construction costs escalate rapidly with depth and again it was felt that the energy savings were not sufficiently significant to warrant the hump profile. In the bored tunnel sections, however, the energy saving profile is compatible with efficient tunnelling procedures. The increased depth of the tunnels gives increased cover to the boring and thus generally greater security for the construction works and the shallow stations minimize the passengers' vertical travel time. The saw-tooth profile was therefore adopted wherever possible with acceleration gradients of 3% and deceleration gradients of 2·5% leading to humps of 6 m nominal value.

11. It was also important to provide a horizontal alignment without speed restrictions to allow an optimum acceleration, constant speed, coasting and deceleration regime to be adopted for all inter-station runs. The criteria which meet this

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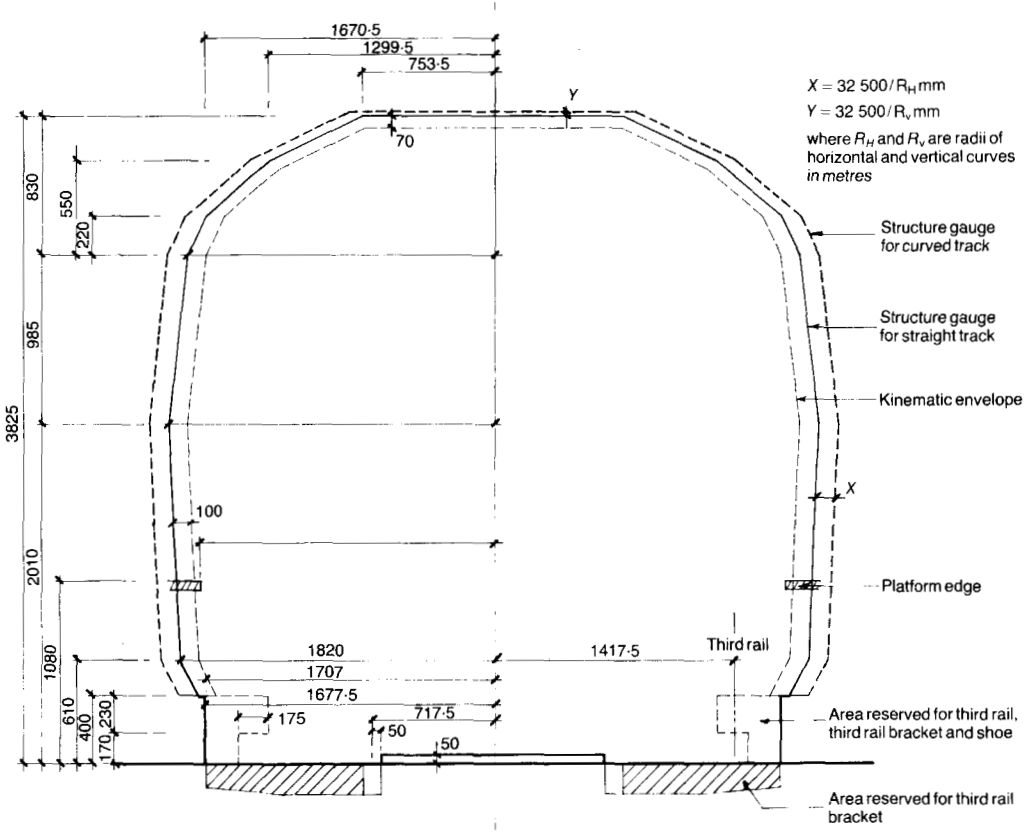


Fig. 2. Structure gauge

requirement also reduce rail and train wear and thus minimize maintenance. Alignment criteria adopted are shown in Table 1. In practice there are only three curves on the East-West line and one curve on the North–South line below 300 m. The minimum curve used on the main line is 275 m. The minimum curve in depots is 190 m.

12. *Platform screen doors.* With the year-round high temperatures and high humidity which prevail in Singapore, air-conditioning of stations, at least in the underground lengths, was inevitable. Although it would have been possible to provide adequate natural ventilation to allow the trains to run without air-conditioning in the open, the savings are doubtful when balanced against the heavy load this throws on to the plant as the heated carriages enter the underground lengths. Similar reasoning led to the adoption of covered sidings which ensure that trains entering service at any time of day are at an acceptable temperature. The Singapore metro is thus the first to be fully air-conditioned throughout the year and environmental energy costs in consequence are high—particularly in relation to traction power.

13. Studies in phase III of the MRT study highlighted the major economies—up to 50% of the operating costs of a conventional system—that could be achieved if platform edge screen doors could be installed to separate the air-conditioned stations from the tunnels. Tunnel ventilation would be adequately provided by train movement under normal circumstances and in the event of stoppages by emergency fans.

14. It was desirable to adopt this concept. However, one remaining unknown was the station door reliability. Train door failure is one of the most significant reasons for service delays in rapid transit systems; however, trains can be taken out of service. The implications to an operator of station door failure are evident: one can hardly take a station out of service. When detail design started therefore it was thought prudent to provide adequate space—in plant rooms, ducts and so on—to allow additional air-conditioning plant to be installed if it became necessary. Should, therefore, the screen door concept prove impractical for whatever reason a fallback to a conventional system is available. This decision was taken pre-tender and the costs associated with it are contained within the lump sum

Table 1. Alignment criteria

Curve radii: m	Minimum 200
	Preferred minimum 400
	Minimum within depots 150
Transition curves	Cubic parabolic form
Gradients	Absolute maximum 3%;
	Between 1% and 3% length to be as short as possible
	Less than 1% length unrestricted
Drainage gradient	Desirable minimum 0·5%
	Absolute minimum 0·25%
Cant: mm	Maximum 150 for concrete track, 125 for ballast track
	Desirable maximum 125 for concrete track, 110 for ballast track
	Maximum deficiency: desirable 90, absolute 110
Cant gradient	Maximum: 1 : 500
	Preferred: 1 : 750
	Minimum: 1 : 100

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price. However, in the context of the overall dimensions of the station box the additional volume required is not great.

Civil defence aspects

15. The decision to strengthen underground stations for civil defence (CD) was taken relatively late (end of 1982). However, five stations—Braddell, Newton, Somerset, City Hall and Raffles Place—were held back and the tender documents were revised to incorporate the CD criteria. All underground stations in subsequent phases have been designed to CD standards i.e. Tiong Bahru (phase IA), Marina Bay, Bugis and Lavender (phase 2A).

16. The civil defence concept envisaged a fully protected shelter area cut off from the outside world by gas- and blast-proof closure devices. To protect these closure devices from direct weapons effect semi-protected entrance areas are provided. Enhanced design criteria particularly for shock loadings are required for the main structure, the architectural finishes and the electrical and mechanical (E & M) equipment within the fully protected zone. The stations are separated from the running tunnels by sliding blast doors. The tunnels can be ventilated by the emergency tunnel fans and could therefore act as secondary shelters offering protection although at a lower level to that afforded by the stations.

Fire protection and life safety

17. Until 1983 there was no officially recognized code that dealt with fire protection and life safety. The concept design of the Singapore system had in earlier stages been based on practice adopted in European and North American systems. Local building codes in Singapore were not readily applicable to a railway system. A committee with representatives from the MRT Corporation, the Singapore Fire Service (the regulatory body for all matters concerning fire prevention) and the Development and Building Control Division of the Public Works Department was set up to formulate fire precaution requirements. Rather than try to develop an original code this committee decided to adopt in toto NFPA 130, the standard issued in May 1983 by the National Fire Protection Association Inc. USA and especially prepared for use in fixed guideway transit systems.

Station layout

18. Having established the train formation, the station layouts followed. The platform length of 138 m (6 × 23 m long car trains) is the main parameter. Operating considerations favoured an island platform layout to allow

(*a*) efficient use of platform width particularly at stations with tidal flow passenger characteristics
(*b*) variable escalator and stair usage according to passenger flow
(*c*) efficient deployment of station staff to supervise platform movements
(*d*) user friendly access to the departure platforms without the need for the passenger to make a prior choice of platform in the booking hall.

The construction implications of island platforms are that

(*a*) bored tunnelling construction favours the concept
(*b*) cut-and-cover techniques incur a cost penalty
(*c*) elevated island stations impose additional curves on the alignment but

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costs marginally favour the concept because of the overall reduction in station width.

19. The decision was taken to adopt island platforms wherever feasible to maximize operating efficiencies and thus minimize running costs. Island platforms are provided at 41 of the 42 stations; the one exception, Braddell, is due to the constraints imposed by adjacent buildings on the tunnel alignment as it approaches the station. At three stations—Ang Mo Kio, Jurong and Tanah Merah—three tracks are provided with two island platforms. These stations are all adjacent to depots or stabling sidings. The third track is to allow the intermediate reversing of services and the introduction of trains from and their return to the depot without disruption to the service.

20. Single ticket halls are provided at all stations which reduces staffing levels and simplifies station control. Traffic flow capacities of stairs, passages escalators and so on were based on recognized data derived from other systems[1].

21. The standards set by NFPA 130, of 4 min for the evacuation of the occupant load from the station platform and 6 min for evacuation from the most remote point on the platform to a point of safety, combined with the code's limitation on the use of escalators in emergency resulted in significant staircase requirements: these have been accommodated generally by open staircases within the platform's length and enclosed flights at either end. At a few stations (Orchard, Dhoby Ghaut) a central feature staircase serves both access and architectural needs. NFPA 130 requires fire compartmentation. Public areas are required to have a 2 h fire resistance separation against non-public areas. Within the non-public areas of the transit zone the fire resistance ratings of partitions vary, being dependent on the risk of each occupancy. Zones of non-transit use (for instance shopping floors) are required to have 3 h separation from areas of transit use.

22. Elevated stations, which required only minimal air-conditioning of non-public areas, were relatively simple in layout. Off-street stations were two level with the upper island platform served by escalators and stairs from a ground level ticket concourse. Apart from communications and relay rooms which were located at platform level, plant rooms were accommodated at the ground level which ensures easy maintenance access. Stations in the road medians have generally three levels: plant rooms at ground level, ticket concourse at first level with direct passenger access by overhead bridge from the road sides and the island platform at second level. The roads were widened where necessary to accommodate this station concept. Where only two levels are provided owing to alignment considerations subway access is provided.

23. The additional space required in underground stations to allow for both operation without platform screen doors and the CD requirements (additional provisions for generators, diesel and domestic water storage, and so on) intensified the design task in the detailed development of the station layout. It also complicated the administration of design and construct 'lump sum' contracts in phase I. To provide a reasonable basis for payment the MRT Corporation determined that the conceptual box of the tender documents should represent the basis of the contract price. Where it was established that efficient design development could not reasonably provide an acceptable layout within the specified box, the increased size was accepted as the basis for additional payment. These variations were exceedingly numerous owing to the multifarious modifications to electrical and mechanical equipment which were introduced throughout the civil construction period and on occasion after structural work was complete.

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Viaduct

24. The introduction of such a dominating feature as nearly 50 km of viaduct into the Singapore scene made it crucial that its appearance should be pleasing as well as functional. An investigation of possible superstructure forms was undertaken and six options were selected for further study

(*a*) in-situ concrete slab
(*b*) precast prestressed beam with in situ concrete top slab
(*c*) double tee prestressed concrete beam
(*d*) cast in situ prestressed concrete box girder
(*e*) precast prestressed concrete box girder
(*f*) composite steel–concrete box girder.

The study examined cost, structural form and aesthetics, span length, speed of construction, suitability for transitions where tracks separate and adaptability for special long spans over roads and rivers. The design concept was required to permit rapid erection to meet the tight project programme and to minimize disruption from the construction.

25. The final selection was precast prestressed concrete box girders, with typically 25 m spans. A box girder form is structurally efficient, is suitable for precasting and can be erected by launching gantry travelling over the previously erected viaduct if simply supported spans are adopted. The relatively heavy concrete box girder simply supported on bearings also assists in the suppression of noise and vibration and reduces the likelihood of their transmission to adjacent buildings. Concrete as a material was preferred partly because the local construction industry was more familiar with it than steel but mainly to minimize the need for continuing protective maintenance.

26. The decision was taken to support each track independently. This was partly for operational reasons—damage due to derailment or other incident is limited to one track. From the construction aspect the size and manoeuverability of double track box girders posed serious handling problems. Moreover the need to split the tracks at the island stations required the facility for erecting single track beams in any event. Where spans significantly in excess of the optimum single span (25 m approximately) were required a matching in situ section continuous over one or more columns was used.

27. A clean horizontal soffit was stipulated from aesthetic considerations and this matched the individual simply supported span concept. The clean lines were preserved at the column crosshead by the use of half joints. On the outside edge of the box girder a curved profile was adopted to give a slender appearance, the curve having a softening shadow effect. At the junction between the curve and the sloping side of the box a small horizontal step was introduced which matched the half joint. The external faces of the simple, generally circular, columns were relieved by vertical grooves which minimized the potential conflict between a plain column and curved beam and also matched the stress relieving open joints in the untensioned cantilevered walkway (Fig. 3).

28. To maintain the harmony of the viaduct profile provision was made where the railway left or entered the central median of a road for the alignment of the road to be adjusted, at significant cost, to allow the flat soffit and circular column configuration to be maintained. In only one instance were portal frames resorted to.

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Fig. 3. Viaduct

Track drainage

29. The annual rainfall in Singapore is of the order of 2500 mm. The rainy season occurs approximately in November, December and January, but torrential storms occur throughout the year. Efficient drainage is therefore essential. Tracks in cuttings were avoided wherever practical, the tunnel box being extended above ground at transitions between underground and viaduct. Only adjacent to Bishan station was it necessary because of topographical constraints to accept a long (about 500 m) cutting. In this case storage sumps with a capacity of 4500 m^3, equivalent to 6 h storage for a 1 in 50 year storm, were provided at the portal and both pump failure and high water level alarms are provided back at the Operations Control Centre to ensure comprehensive monitoring and protection for the tunnels against flooding. On the viaduct it was considered acceptable to allow the rainfall generally to drain through the ballast to 150 mm dia. drain pipes at each column. The protective screed to the waterproofing of the box girder was graded to facilitate the flow.

30. As a further precaution overflow pipes were cast, at 7 m centres, into the sidewall of the ballast retaining trough, level with the underside of the sleepers. These are required to allow for any short-term peak storm intensity that exeeds the capacity of the ballast's draining properties, to ensure that the integrity of the electrical resistance to earth and between running rails is maintained at all times. They are set flush with the inner face of the trough, to avoid interference with tamping.

31. Sumps are placed at all low points in tunnels with capacities related to the anticipated water inflow. Very low inflow rates were specified partly to ensure good service conditions for train operation but mainly to avoid the possibility of the tunnels acting as drains and lowering the general water table. The sensitive soils found in Singapore would have exhibited widespread settlement effects if any permanent lowering of the water table had been permitted.

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Contract form

Design and construct

32. The below-ground section of phase I (Bishan to Outram Park) was constructed under 12 design and construct contracts. The contract documents included common design criteria and performance specifications, a particular specification which spelt out the design requirements for each contract and a set of drawings which gave the horizontal and vertical alignment of the track and the conceptual design of the stations. No material and workmanship specification was included in the contract documents. The MRT Corporation considered that the designer himself, as he was taking full responsibility for the design, should prepare the material and workmanship specification to meet the general requirements of the performance specification, but this had the disadvantage of giving rise to a proliferation of different standards and details.

33. The design and construct contracts required designers to submit their designs at three points in time after contract award

(*a*) an intermediate design, when designs were about 70% complete, between 26 and 40 weeks after award

(*b*) a pre-final design, when designs were 100% complete at approximately 12 weeks after intermediate design

(*c*) the final design and contract documentation, when designs were 100% complete and incorporated the MRT Corporation's review comments on the pre-final design, at approximately 18 weeks after pre-final design.

Not only did the contractor's designers prepare the designs in the three stages but to expedite construction progress they submitted numerous separate designs of various items. For a typical contract which included one station and one tunnel section, a total of 500 design packages were received for review.

34. A period of 30 days was specified for review. Design review was managed by one engineer per contract (civil design co-ordinator) within the MRT Corporation whose duty was to ensure that all interested parties were sent the designs, that comments were received back promptly and that a comprehensive and co-ordinated response was returned to the contractor. A careful scrutiny was necessary to ensure that no comment was an instruction to the contractor that compromised his professional responsibility as designer.

35. One section within the MRT Corporation that was sent designs for review was the Building Regulations Unit. The Corporation, being a statutory board under the Ministry of Communications and Information, was exempt from the normal development and building control procedures imposed on private developers. However, in order to ensure that all building regulations were observed, the Corporation set up its own Building Regulations Unit which vetted all designs sent in both by contractors and consultants.

Engineer design

36. All the elevated portions of the railway and the below ground portions on phases IA and II were designed by consultants under contracts appointing them as architect/engineer after a review of priced design services submissions. The main depot and the permanent way serve the project system wide and were therefore also designed by specialist consultants. Seven contracts were awarded (Table 2).

37. Designs were submitted in three stages within the following typical time scale

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Table 2. Design consultants

Phase	Contract	Architect/Engineer	Extent
I	101D	Electrowatt	Main Depot
I	102D	Mott, Hay & Anderson	3·2 km of viaduct 2 elevated stations
IIA	200D	Mott, Hay & Anderson	8·4 km of viaduct 0·9 km of tunnel 5 elevated stations 1 underground CD station
IIA	300D	De Leuw Cather Int'l Wilber Smith & Associates RSP Architects, Planners & Engineers	15·1 km of viaduct 4·0 km of tunnel 10 elevated stations 3 underground CD stations Sub-depot
IIB	400D	Parsons Brinckerhoff/ Maunsell Consultants	16·8 km of viaduct 8 elevated stations Stabling sidings.
IIC	500D	Parsons Brinckerhoff/ Maunsell Consultants	1·5 km of viaduct 1 elevated station
System-wide	110D	Electrowatt	Permanent way

(*a*) the preliminary design at 6 months after contract award when design is typically 70% complete
(*b*) the pre-final design at 9 months after contract award, 100% complete
(*c*) the final design at 11 months after contract award; 100% complete and incorporating MRT Corporation's comments on the pre-final design.

A design review process identical to that for the design and construct contracts was undertaken.

Permanent way

System criteria

38. Standard gauge (1435 mm) was adopted owing to its almost universal use which allows equipment to be specified and obtained by worldwide competitive tender. The criterion for rail design was a 16 t axle load. Structural design was satisfied by a section such as the Union Internationale des Chemins de fer UIC 54. However, from reasons of longer service life, better electrical conductivity because of increased cross-section and greater track frame rigidity improving stability, the UIC 60 section was adopted throughout. This allows the exchange of permanent way material in different stages of wear between main lines and depots and reduces the amount of spares to be kept in stock. In curves with radii smaller than 500 m, in order to reduce rail wear, high wear resistant steel grade 110 with tensile strength of 1080 N/mm^2 and about 1% chrome content was selected, whereas for all other lines grade 90 rails with a tensile strength of 880 N/mm^2 were adopted.

39. To ensure reasonable wear of both wheels and rails the latter were inclined 1:40 in plain track and 1: ∞ in turnouts. For MRT Corporation vehicles the wheel profile ORE-S 1002 was adopted which has been especially designed to match the UIC 60 rail section (Fig. 4).

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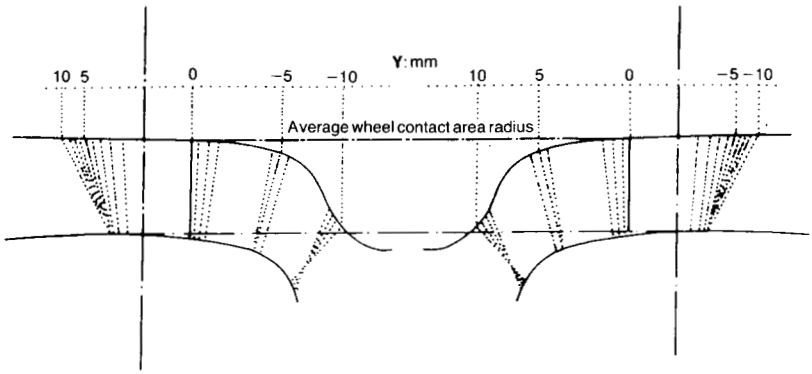


Fig. 4. Wheel–rail contact points

40. The Pandrol system was accepted as a compliant rail fastening system and is used throughout. The clip type with its toe load of about 1000 kgf at 12 mm deflextion is also installed in turnouts and designed to be compatible with the flexible inner stock rail bracing installed at the stock rail assembly within the movable length of the tongue rail.

Ballasted track

41. Ballasted track, installed in all at-grade sections and on viaducts has the advantages of being relatively inexpensive, insensitive to differential settlements at viaduct columns, having good electrical insulation properties and allowing easy maintenance of cross levels, alignment and regrading at soft ground areas. A major consideration in the selection of ballasted track for all open sections was that when combined with wooden sleepers the sound level is as much as 3–5 dB(A) lower than that of slab tracks. The total viaduct dead load is increased only by approximately 1% by the use of ballast track as against the fixed slab track system used in the tunnels.

42. Ballast was supplied from the Karimun Quarry in Indonesia as the only local sources capable of meeting the strength requirements of the Specification were committed to concrete aggregate production. Quality was ensured by the prior inspection and testing of faces at the quarry to determine their suitability, intermittent checks on the quarry production, acceptance in Singapore only against agreed samples and regular sampling and laboratory testing of deliveries before incorporation in the works.

43. Wooden sleepers were selected as they are easier to handle owing to their light weight, are readily available in the region and can accommodate quite easily the third rail brackets as well as the guide rail features. When correctly impregnated their life is considered equal to that of concrete sleepers. Continuous welded rail was used throughout the system.

44. On viaduct sections the UIC 60 rails are installed on base plates fixed to sleepers at 700 mm centres. As a protection against derailment two additional rails are installed between the running rails as derailment guards (Fig. 5). This was

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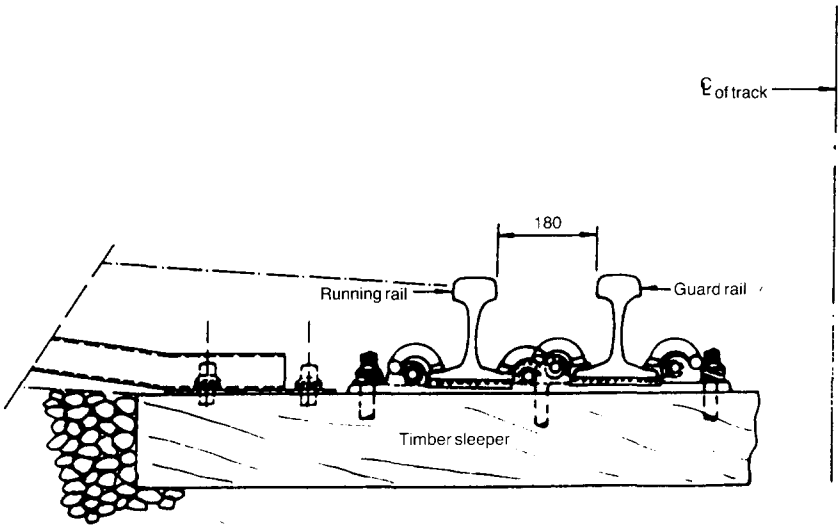


Fig. 5. Guard rail assembly

adopted as being a more positive measure than for instance a concrete upstand. Although expensive in initial cost the facility for reuse as running rails improved the economies of the system. The derailment guards are supported on every second sleeper on a common base plate; they are insulated from the running rails and discontinuous every 18 m to avoid interference with the signalling circuits.

45. Expansion devices were not considered necessary over the free bearing ends of viaduct beams. The temperature range in Singapore is relatively small—ambient temperatures vary only from 24 to 36°C throughout the year and the differential movement between rail and structure was accommodated easily by the ballast bed. Even though calculations showed that a differential movement of up to 10 mm could be accommodated without any damage to the track, the maximum differential movement due to column sway and so on was limited to 5 mm in the design criteria.

46. The standard base plate is fixed to the sleeper by three SS8 screw spikes whereas the guard rail base plate is fitted with five screw spikes, holding the base plate in place elastically with double spring washers. A 4·5 mm thick high density polyethylene pad is installed between the rail and the base plate.

47. Kempas sleepers of 2600 mm length 260 mm width and 160 mm height from neighbouring Malaysia and Indonesia are used. This wood had been selected over hardwood species like oak for its superior absorption characteristics. It absorbs up to 150 kg creosote per cubic metre achieved by a full cell pressure–vacuum impregnation process which protects the wood well against fungus and insect attack. Experience with sleepers made from this wood species has been gained in the neighbouring countries.

48. The sleepers are embedded in ballast grade 20–65 of 1800 kg/mm^2 minimum compressive strength. The design ballast depth at the rail location of 300

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mm below the sleeper is maintained to ensure an overlapping of the ballast pressure zones. For at-grade sections and at depots a 500 mm thick blanket of impermeable sub-ballast is placed to ensure a stable track support and a high run off surface for rainwater (Fig. 6).

Fixed track

49. For tunnel sections where airborne noise does not play a dominant role and substantial settlement and differential movements under trains are not expected, slab track was the obvious choice. It provides a lasting alignment in the narrowly confined tunnel envelope, requires minimum maintenance and guide rails are not installed. It also makes drainage of the tunnel floor easier and the floor level can be kept clean with less effort. Cables and services crossings are cast in situ. Additionally the construction height of slab track compared with ballasted track is reduced by 120 mm (Fig. 7).

50. Out of the variety of systems available the one with the track frames cast into the slab has been adopted. It provides the advantage that the whole frame can be completely assembled to design gauge, line and level before casting in. No further adjustments are required and the track positioning achieved is excellent.

51. The rails are insulated from the base plate by a rail pad and Pandrol insulators fastened with two Pandrol clips to the base plate. The base plate is secured by two 33 mm dia. high tensile bolts cast into the sleepers. Resilience is provided by a 14 mm thick elastomeric rail pad. The compression springs which accommodate the deflexion are kept in place by selflocking hexagon nuts (Fig. 8).

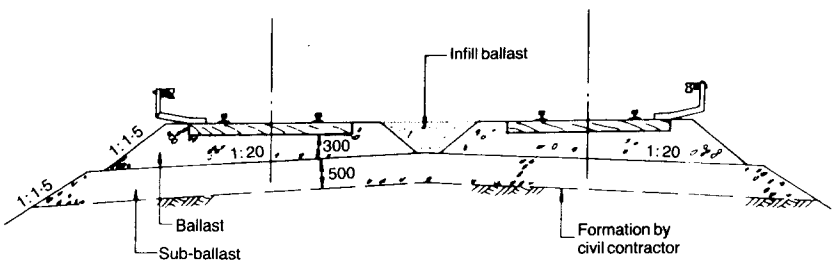


Fig. 6. Cross-section of ballasted track

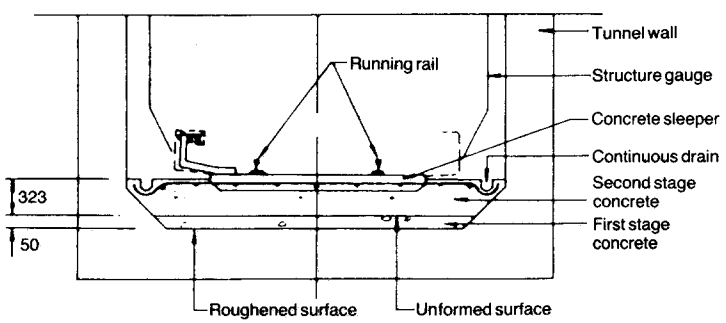


Fig. 7. Cross-section of slab track (cut-and-cover tunnel)

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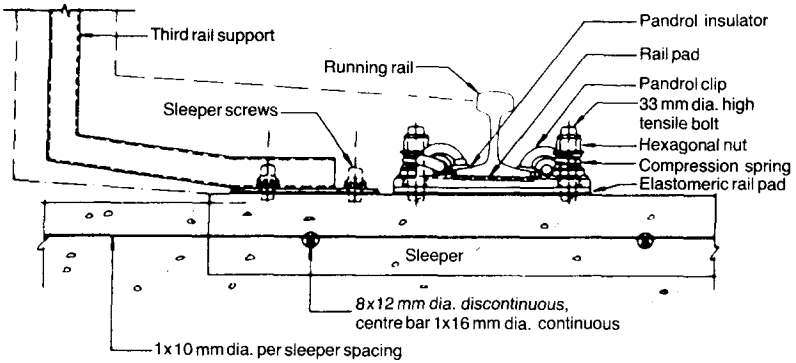


Fig. 8. Slab track fastening

52. The sleepers are of concrete grade 55, prestressed with eight 5 mm dia. indented wires and are cast into concrete grade 35. Bonding of the sleepers to the second stage concrete is improved by roughening their sides and by reinforcement passing through 50 mm cast-in pipes. Successive sleeper bays were concreted to full height to ensure the hydraulic head was adequate to force the concrete through the 50 mm pipes around the reinforcing bar. Aggregate was 20 mm maximum size.

Floating slab track

53. Sections of the route cross densely populated areas and commercial districts where disturbance due to vibrations and secondary airborne noise inside buildings is unacceptable. For these sections the single unit type of floating slab track (FST) as used in Hamburg and Munich was selected. The unit weighs between 3 and 4 t and can be transported easily into the tunnel as it is only 690 mm wide. Special lifting hooks have been provided to facilitate maintenance of the drainage under the FST system. As the individual units rest on three bearings, the permissible tolerances can be relatively large and are made good during installation of the track. The track frame is preassembled to gauge, line and level and then cast-in at each individual FST unit where special recesses allow embedding of the sleepers. This results in finite tolerances being achieved for the finished track, as all cumulative tolerances are made good before placing the concrete.

54. The mass of the FST per metre of track was designed to be greater than that of the train, and the bearings deflect under load similarily to the ballasted and slab tracks, resulting in a natural frequency well below 16 Hz so that any transmitted noise is a relatively acceptable rumble. Experience has confirmed that the FST effectively reduces both vibrations and secondary airborne noise such that train operation is imperceptible in adjacent buildings. However, in the tunnels the noise level from the wheel rail interface is significantly increased (by 6 dBA approximately) and this when allied to the increased cost is a major deterrent to the extensive use of FST. For the FST sections the rail rests on high density polyethylene pads and is fitted by Pandrol rail clips and Pandrol shoulders cast in prestressed concrete sleepers otherwise identical to the ones used for slab track sections (Fig. 9).

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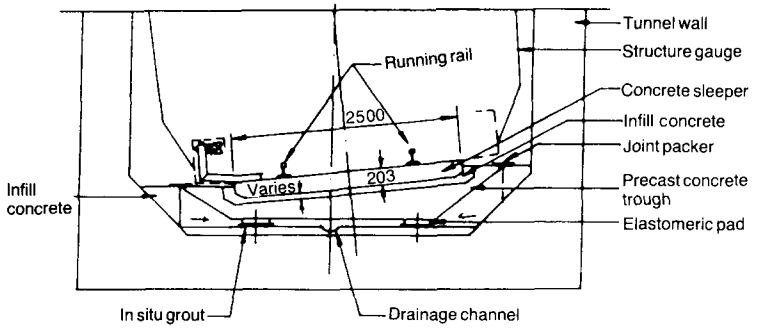


Fig. 9. Cross-section of floating slab track (curved track)

55. The sleepers are cast into the FST units with infill concrete grade 35 with expansion additives. The FST units themselves are precast elements 690 mm wide 450 m high approximately and vary in length to suit the tunnel cross-sections. They are formed in reinforced concrete grade 35. Three different shapes of units are cast to allow for the rail superelevation around curves. The gaps both between units and between the units and tunnel walls are covered by plastic joint packers to avoid foreign particles falling in between and creating a bridging effect (Fig. 10). The passage of the rolling loads across the individually supported units causes stress reversals in the infill concrete but enquiries made of similar systems (Hamburg, Munich) indicated that no serious consequences have been experienced over five years' use in revenue service. In Singapore the use of an overdesigned rail section (suitable for 16 t axle load) will, it is hoped, minimize this effect under the relatively light (12–14 t) axle loads imposed in practice. No signs of distress have been observed after one year of revenue service.

Turnouts

56. Most of the 258 turnouts installed (181) are of the type 190–1 : 9 (i.e. a centre line radius of 190 m and a deflexion angle of 1 : 9 (6°34")). A further 41 of

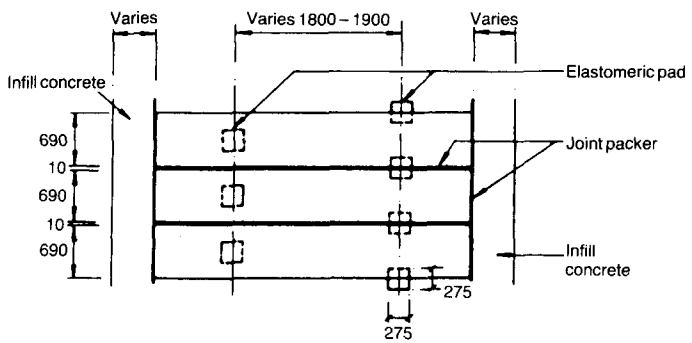


Fig. 10. Arrangement of FST units

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type 300–1 : 9, 28 of type 500–1 : 14 and 8 of type 190–1 : 7·5 are installed. Turnouts for all three track support systems are identical in construction with the exception of the base plates. Those are modified to suit either the sleeper screws or the holding down bolts.

57. For ballasted track sections the turnout is placed on wooden sleepers of 260 × 160 mm cross-section, with varying lengths to suit their position within the turnout. In the FST sections the turnout is supported on a composite slab. A 80 mm prefabricated filligre slab is placed on the bearings. This is dimensioned to flex under the load of the first 100 mm layer of the superincumbent in situ concrete in order to equalize the manufacturing and installation tolerances of the bearings. The second, 450 mm, in situ concrete layer is then placed to bring the slab up to full height. For slab track and floating slab track sections the turnout holding down bolts are cast into the concrete after holes have been drilled at the exact location through templates. The top reinforcement layer is placed sufficiently low in the slab to avoid damage or interference with the drilling.

58. In areas sensitive to noise and vibration and at emergency crossovers swing nose crossing turnouts (SNC) are installed. The SNC itself consists of the fixed wing rails, the dilatation joint and the movable solid block nose. During the operation of the SNC, the nose is moved to completely close the gap at the nose–wing rail wheel transfer (Fig. 11), resulting in reduction in noise and vibration and increase in service life of the material in contrast to the fixed common crossing (Fig. 12).

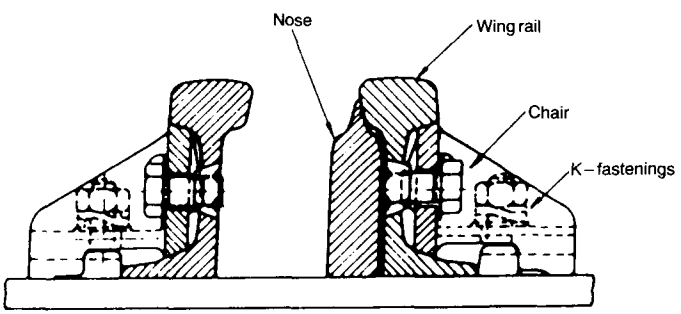


Fig. 11. Swing nose crossing (section at nose)

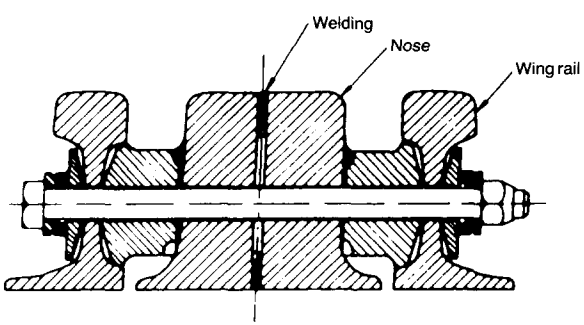


Fig. 12. Common crossing (section at nose)

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Third rail system

59. An under-contact aluminium–stainless steel composite conductor rail was adopted. The under-contact has the advantage that dust and dirt does not accumulate at the contact surface where it may lead to excessive rail and collector shoe wear. Insulation is provided by a relatively small cover, protecting both the passers-by from the live rail and the conductor rail from accidental damage and dirt. The covers are provided with a 35 mm gap around the rail for air circulation and electrical clearance.

60. The relatively light weight of the aluminium rail of about 16 kg/m allows the third rail supports to be spaced at up to 6·3 m distance and to be of light construction. The rail's low resistance of about 7 mΩ/km reduces the voltage drop and allows a greater spacing of substations. The cross-section of the rail of 5200 mm^2 with a 4·5 mm thick stainless steel contact strip has a permissible continuous current carrying capacity of 4700 A at 40°C. The rails are of 15 m length, connected by Huck bolts and fishplates. Every 90 m outside tunnels and 180 m in tunnel sections an expansion joint caters for the longitudinal expansion of the rail, which is held in position at the mid point by anchors. At turnouts, walkways, road crossings and other gaps in the third rail ramps are provided, inclined 1 : 50 m at main lines and 1 : 30 at depots. The gaps are bridged by 400 mm^2 single core copper cables laid in troughs. The conductor rail is fitted at a height of 170 mm above and at 800 mm distance from the running rail on glass fibre reinforced polyester insulators with claws bolted on to rectangular galvanized steel section supports mounted on to the sleepers (Fig. 13).

Ground conditions

61. The geology of Singapore is complex with the predominant Bukit Timah granite and Jurong Formation of sandstone, siltstone and mudstone traversed by river valleys infilled with very soft marine clays and fluvial silts. The granitic and

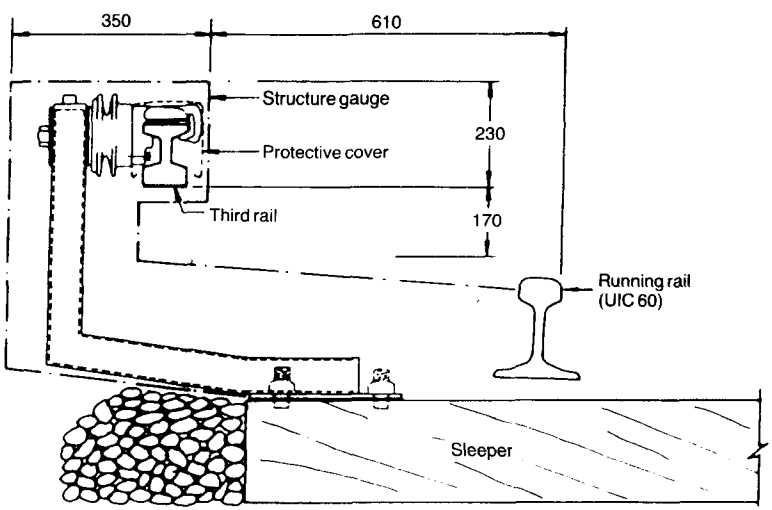


Fig. 13. Third rail support assembly

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sedimentary rocks have suffered deep tropical weathering, which has decomposed the rock to a medium to stiff silt or clay.

62. A major problem encountered in the design was the determination of an at-rest coefficient K_0 for the design of cut-and-cover tunnels and station structures in the completely weathered rock. In the pre-historic past when the material was unweathered, it had undergone folding and faulting and was put into high horizontal stress. It is likely that much of the over-consolidation was destroyed by the weathering process but it was doubted whether the conventional $1 - \sin \phi$ coefficient, typically 0·5, would be suitable in such an un-deposited clay. It was concluded that a value of $K_0 = 0{\cdot}8$ to 1·0 was more realistic.

63. The recent deposits of the Kallang formation, which comprise very soft clays, sometimes with organic deposits, are highly compressible and not suitable as a founding stratum. Where these deposits are deep the station foundations (diaphragm walls, piles or barrettes) have been extended down to an underlying bearing stratum. The flexible bored tunnels were allowed to float in the soft marine clay, but the cut-and-cover tunnels were supported on diaphragm walls or piles if there was marine clay under the base slab. At Marina Bay the possibility of allowing cut-and-cover tunnels to float in the marine clay (C_u 15 kN/m^2 minimum) was considered because of the depth and cost of the piling needed. However, in this case allowance had to be made for the placing of 8 m of fill over the tunnels during future reclamation work. A settlement analysis was carried out and a suitable detail for a tunnel joint designed. The track within the tunnel would have needed to be ballasted to allow for future adjustment and the height of tunnel made adequate for this purpose also. However, there was concern about the confidence which could be placed in the settlement analysis, the possibility of transverse rotation, and the ability of the joints to withstand the movement. Furthermore, there was the very significant ongoing maintenance cost of tamping and reballasting of the track to consider. The MRT Corporation finally opted against the proposal and the tunnel has been designed supported on piles.

Tunnels

Criteria

64. The criteria given to the designers were as follows.

(*a*) Tunnel size was 5·2 m i.d. minimum. This figure included an allowance of 100 mm on radius for inaccuracies due to manufacture or construction.
(*b*) Segmental lining was to be of reinforced concrete.
(*c*) A plane strain continuum design method was to be used based on an elliptical deformation as proposed by Muir Wood[2] and later supplemented by Curtis[3] to include radial deformation due to tangential stress in the ground.
(*d*) Maximum deflexion under load was of 25 mm on radius.
(*e*) Watertightness was to be 5 ml/m^2/h overall; 10 ml/m^2/h for any 10 m length of tunnel.

All contractors opted for a tunnel size greater than 5·2 m, the selected sizes ranging from 5·23 m to 5·4 m. Even at 5·4 m several realignments became necessary during tunnel driving due to out-of-tolerance construction, usually in the vertical direction.

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Tunnel stability

65. In the preliminary design stage, one of the principal concerns of the MRT Corporation was the stability of bored tunnel linings in very soft marine clays. There are many locations along the alignment where the tunnel is driven in a full face of the marine clay.

66. The instability problem that was of particular concern was the ability of the clay to withstand the quasi-flotation effect of the tunnel. A finite difference elastic-plastic model with a Mohr-Coulomb failure criterion was set up which provided a non-linear large strain capability. Many cases were considered to determine the critical uplift condition including different arrangements of excavation (i.e. canals) and surcharge at ground level and conditions of total and effective stress. The conclusion was that there was no case of instability with a clay shear strength over 5 kN/m^2; in most conditions 3 kN/m^2 was adequate. With the Singapore marine clay having a minimum shear strength of, typically, 15 kN/m^2, it was concluded that the tunnel was safe against uplift under all conditions.

Segmental lining

67. *Flexibility.* All tunnel designers needed to make a design decision on the flexibility of the lining where the tunnel is located in areas of soft marine clay. The marine clay is sensitive and there was concern that the tunnels would deflect more than specified unless they were of adequate stiffness. The designers all decided to provide a relatively stiff lining with staggered longitudinal joints, a full bearing area on this joint and only five or six segments, excluding the key, per ring. One designer made a detailed analysis of how a staggered jointed lining behaves and determined how much moment in the segment could be transmitted across the stagger to adjacent segments and how much the joint itself should transfer. All designers were faced with similar axial thrust and had opted for similar flexibility and hence design moment, but their particular solutions in terms of reinforcement detailing, bolting arrangments and waterproofing methods differed markely (Fig. 14).

68. *Bolting.* Steel bolt pockets in the circumferential joint minimized the amount of concrete loss from the cross-sectional area of the segment body, but prevented a strong edge flange being formed in the segment and thus weakened the segment with respect to resistance to shield jacking loads. Concrete bolt pockets of contracts 106 and 109 have the reverse effect, as they allow a strong edge flange but a substantial part of the cross-section is lost. The best arrangement is the curved bolts of contracts 107B and 301 or the dowels of contract 107. Steel bolt pockets in the radial joint were found to be without disadvantage, making the segments easy to build accurately.

69. *Waterproofing.* Waterproofing suffered from the decision to allow the contractors to write their own material and workmanship specification. In all cases, designers elected to provide a sealing material in grooves near the rear edge of both the circumferential and radial joints as the main line of defence against water ingress. A variety of materials were used for sealing the linings but, with one exception, they did not perform to expectations. The only successful sealing material, and markedly so, was a non-water soluble but water expansive resin and synthetic rubber product. When the material comes into contact with water, it swells up to ten times its original volume, filling the joint gap and sealing against water ingress under its own expansive pressure. As the material may be fitted flush within the groove, few problems of damage or misalignment, even during key installation, occurred.

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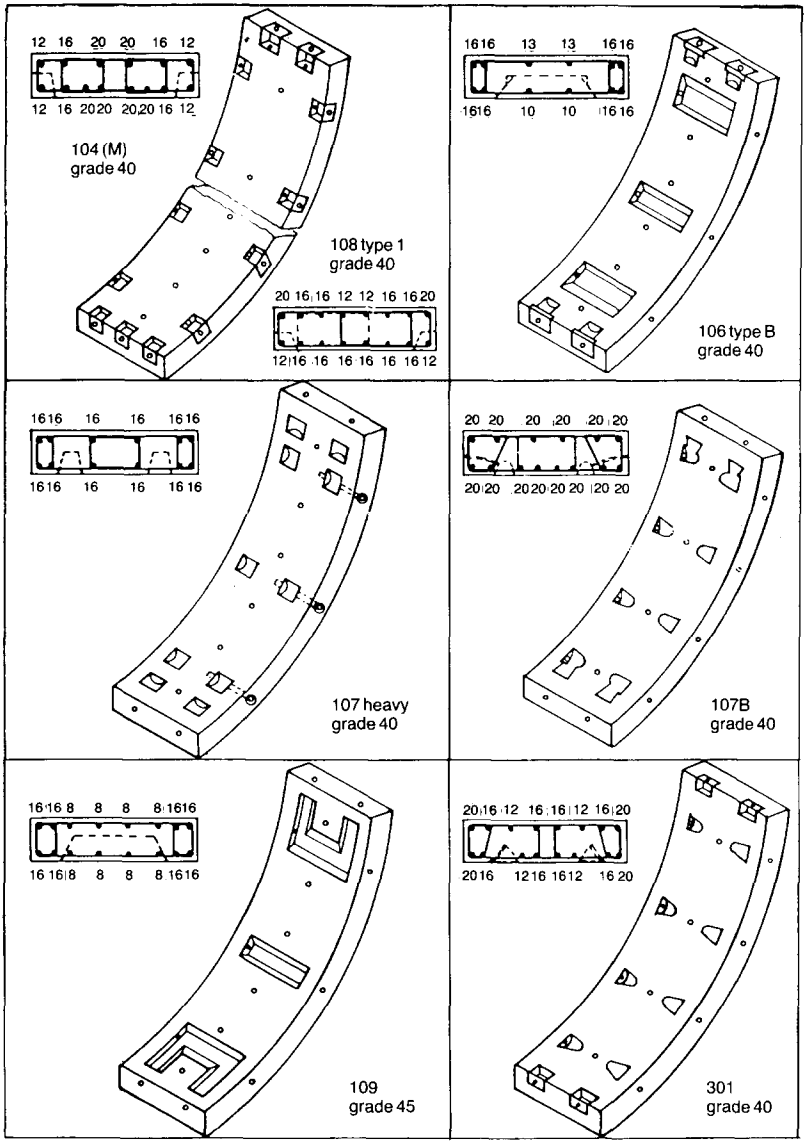


Fig. 14. Segment details

70. *NATM.* NATM tunnels were designed for sections of contracts 104, 105, 107A and 107B. Although generally utilized in better ground, i.e. the partially or completely decomposed sandstones or granites, it has also been carried out in chemically treated sand and coral beds. Rock bolts and shotcrete have been employed in the soft rock with shotcrete thickness up to 200 mm whereas ribs and

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shotcrete have been employed in the softer grounds. A minimum lining thickness of 400 mm of in situ concrete was specified to ensure proper placement and compaction. Nominal reinforcement was adopted for crack control.

71. The MRT Corporation took the decision to incorporate silica fume into the concrete mix to provide a more impermeable lining. After initial reluctance from the contractors, the other advantages of the use of silica fume became apparent. A mix with fume added at 10% by weight of cement produces a workable sticky mix, easy to pump with a marked reduction in bleeding and segregation. A reduction in permeability of 10 to 20 times was achieved.

72. *Temporary lining.* A temporary lining of expanded ribs and lagging was used for most of contract 105 in mixed ground conditions ranging from shattered rock to marine clay (Fig. 15). The laggings, 125 mm wide, fitted snugly between the flanges of the ribs. The lining was built in the tailskin and then shoved clear. A sheet of plywood was nailed to the back of one piece of lagging and when jacking to expand the rib took place, the gap in the lagging exposed the plywood rather than the ground.

73. The lining was designed on the assumption that arching of the ground between ribs would reduce the loading on the lagging. Shield thrust load on the lagging was also considered. Ribs were designed for full overburden.

74. Problems arose from water ingress in the more permeable sections of decomposed granite and in the fluvial silts and sands. Compressed air would be

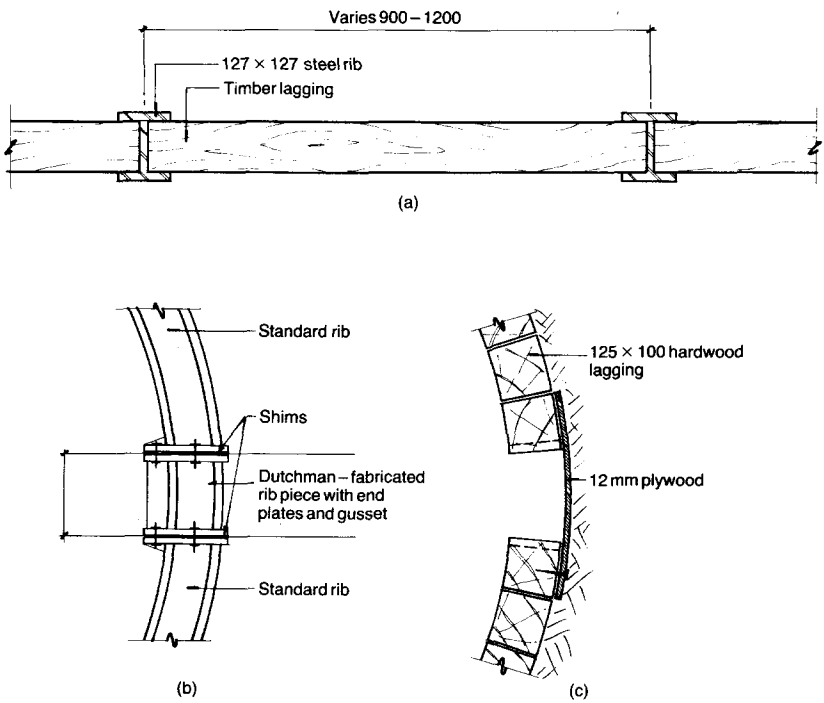


Fig. 15. C105 expanding ribs and lagging: (a) section; (b) rib; (c) lagging

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determined by the requirements at the face which at times was not sufficient to hold back water in the full length of tunnel drive. The compressed air was removed before the placing of the primary lining. This led to substantial water ingress and consolidation settlement in some sections of the marine clay length. In the interim between decompression and permanent lining up to 100 mm of distortion to the ribs took place and intermediate ribs were inserted to contain the movement.

Groundwater quality

75. One of the first problems identified by the MRT Corporation that had not been fully appreciated by the designers was the presence of acids in the ground. Most of the soil types appeared to exhibit high acidity in places, producing pH values well below the minimum of 6·0 recommended for ordinary portland cement. Where pH was low, sulphates were found to be high. The areas appeared to occur in local pockets often with substantial lengths of neutral ground conditions in between. No correlation between these acidic pockets and any surface or geological feature could be established. One disconcerting fact was that samples taken by the contractor and the MRT Corporation from the same source often gave different readings. Furthermore, litmus paper tests in situ did not show high acidity to anything like the levels of the results coming from the laboratories.

76. Bacterial action, principally aerobic (i.e. chemical reaction that only takes place in the presence of oxygen) was determined to be the cause. The laboratories and site were getting different values because the soil was being exposed to air for different lengths of time. All the tunnelling works are below the water table and therefore aerobic action will cease once the tunnel is built and the water table has recovered. Nevertheless, most of the bored tunnel segments have been coated on the outside with coal tar epoxy. The additional thickness of concrete and the silica fume additive are considered to be adequate precautions for the in situ linings.

Cut-and-cover structures

Structure types

77. Eleven out of the 15 underground stations were constructed under phase I design and construct contracts whereas the other four stations were engineer's design under phases IA and II. Table 3 identifies some main features of their construction by cut-and-cover methods. Nine underground stations, including the four engineer designed stations, were designed for civil defence requirements. The cross-over tunnels located next to Toa Payoh, Newton, Raffles Place, Outram Park, Bugis and Marina Bay stations and the transition structures connecting underground tunnels to viaducts located north of Braddell station, east of Lavender station and west of Tiong Bahru station were designed as cut-and-cover box structures. In addition tunnels under the Singapore River and the Telok Ayer Basin were designed as cut-and-cover tunnels. In total 8·4 route km of underground structures were built.

78. Generally underground stations have two levels—concourse and platform. However, in Orchard, Dhoby Ghaut, Tanjong Pagar and Raffles Place stations an additional shopping level was provided. Raffles Place station also has an additional platform level to accommodate both the north–south and east–west tracks and is the largest in the whole system (Fig. 16). City Hall station also has two platform levels to accommodate the tracks of both lines.

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Table 3. Station construction

Station	Type of contract	Type of station	Number of levels below ground	Depth of excavation	Type of cofferdam employed	Other main features
Bradell	Design and construct	Civil defence(CD)	2	14·9	Diaphragm wall	Side platform due to site constraints
Toa Payoh		Non-CD	2	13·5	Sheet pile	Bus interchange and commercial building built on top
Novena		Non-CD	2	14·7	Sheet pile	Future high rise building planned on top
Newton		CD	2	14·3	Diaphragm wall	First CD station built
Orchard		Non-CD	4	21·0	Bored pile wall + nailed slope	Future development planned on top
Somerset		CD	2	16·2	Diaphragm wall + sheet piles	
Dhoby Ghaut		Non-CD	2	16·1	Sheet piles + shotcrete	Future high rise building planned on top
City Hall		CD	4	22·3	Sheet pile + ground anchors	
Raffles Place		CD	4	25·2	Soldier piles + timber lagging + ground anchors	Largest station
Tanjong Pagar		Non-CD	2	17·9	Open cut + soldier piles + ground anchors	Future high rise building planned on top
Outram Park		Non-CD	2	13·9	Soldier piles + sheet piles	
Tiong Bahru	Engineer's design	CD	2	14·1	Soldier piles + shotcrete	
Bugis		CD	2	18·3	Diaphragm wall	Future high rise development planned
Lavender		CD	2	16·5	Diaphragm wall	
Marina Bay		CD	2	16·4	Flooded sheet pile + tremie slab	

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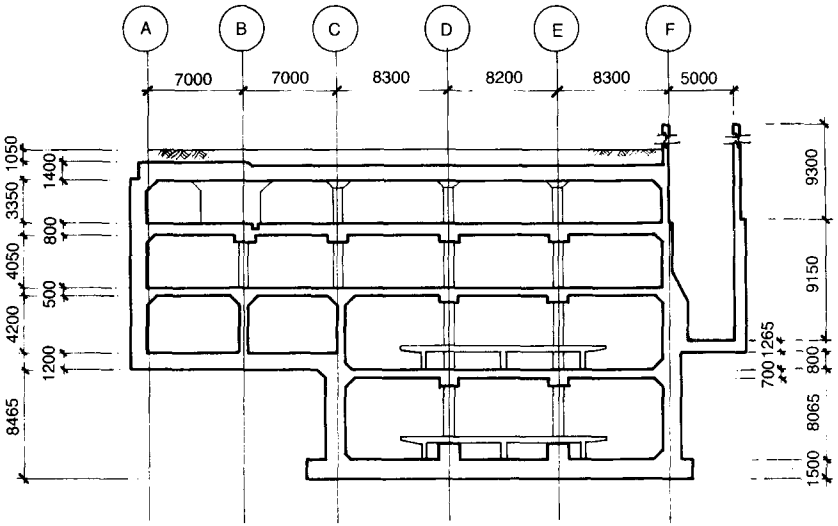


Fig. 16. Cross-section of Raffles Place station

Design requirements

79. Structural design requirements were based on CP110. However, because underground structures need special considerations which are not within the scope of CP110 some guidelines were derived based on BS5337—water retaining structures. Railway design loads were based on BS5400.

80. Where there were roads above, the underground structure was designed for the 45 units of HB surcharge loading given in BS5400. However, the HB loading was placed within the central 5 m of the carriageway and no HA loading was considered together with HB loading. Where the cover under a roadway to the top of the roof slab was more than 1·5 m the HB loading was replaced by a 15 kN/m^2 UDL load. For all underground structures a 15 kN/m^2 surcharge UDL at ground level beside the structure was considered. Surcharge loads from adjacent structures were considered where applicable.

Protection against corrosion

81. Nominal cover to reinforcement was increased to 75 mm where aggressive soil conditions were present. In particular for the cut-and-cover tunnels between Bishan and Braddell stations where estuarine deposits were encountered, 75 mm nominal cover and SRC cement was specified owing to the soluble sulphates and acidic conditions which were present. At cut-and-cover tunnels adjacent to the Singapore River various additional measures were taken over and above the 75 mm nominal cover to allow for the high chloride content in the water (as high as 17 000 ppm) and the suspected acidic soil properties. They include coal tar epoxy coating, additional shotcrete and reduction in w/c ratio to 0·45.

82. In crack width calculations the maximum crack width was limited to 0·2 mm or 0·004 times the nominal concrete cover whichever was the lesser. A minimum 0·125% crack control reinforcement in each direction at each face was specified for all concrete slabs and walls of 225 mm thickness and over.

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83. For thick concrete sections, the maximum allowable core temperature was limited to 80°C and the temperature differential was limited to 20°C. In every contract, these values were monitored for the first few concrete pours and acceptable formwork striking times and curing periods were agreed with the contractor based on the monitoring results.

Construction sequence

84. Particular attention was paid to the construction sequence and, where applicable, to the locked-in moments created on partially completed structures by temporary support loads. In Marina Bay station the contractor proceeded to construct the main walls and roof slab before constructing the concourse slab. In Novena, Toa Payoh and Marina Bay stations partially completed walls which were acting as cantilevers were subject to lateral soil pressure from sheet pile walls owing to strut removal sequences. In such cases additional reinforcement was added to cater for the additional locked-in moments. In cut-and-cover tunnels where a section of the tunnel was completed and backfilled before construction of the adjacent section, details were developed to ensure that the connections between the adjacent sections were made only after the later section also was completed and backfilled (Fig. 17). Similar consideration was given to connections between cut-and-cover box tunnels and stations where a sudden change in backfill surcharge load occurs due to the tunnel roof level being much lower than that of the station roof.

Movement joints

85. Movement joints across the structures were avoided everywhere except for Orchard station where the designer choose to provide movement joints to allow for the construction sequence (both end sections of the station were constructed by the top down method to allow an early start of tunnelling while the centre was constructed from the bottom up) and varying future development loads. A stringent 0·3 mm differential settlement limit between adjacent sleepers specified by the permanent way designers (beyond which adjustment to track components

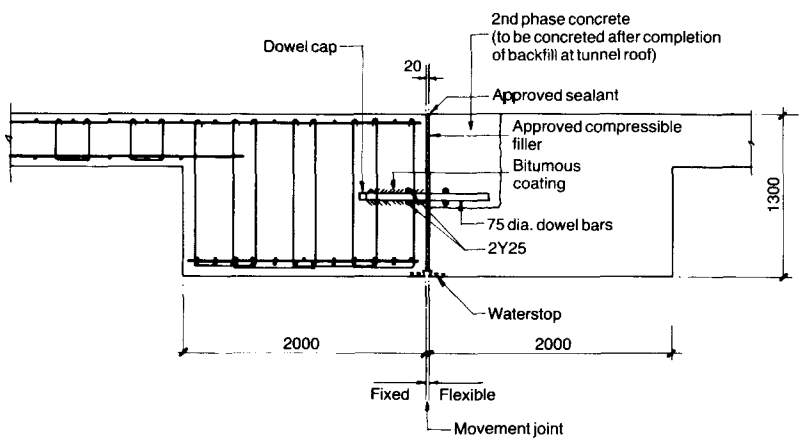


Fig. 17. Connection between cut-and-cover tunnel base slabs

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becomes necessary) made movement joints virtually unacceptable in trackway. However, rotation joints with continuous base slab and discontinuous walls and roof were provided where changes in foundation or differing soil conditions were present.

Design against flotation

86. Two conditions of flotation were considered: one with full backfill load placed on the roof and the second with 1·5 m of backfill material removed above the structure. In the former case a factor of safety of 1·2 was required. In the latter case a reduced factor of safety of 1·0 was considered adequate. In assessing the self weight of the structure, the weights of partition walls, floor finishes, ceiling weight and so on were ignored and only the structural concrete and first stage track concrete weight was considered with a partial safety factor of 1·05. Weight of backfill material was considered with a partial safety factor 1·3 and shear resistance of soil below 2 m from ground surface was considered with a partial safety factor of 2·0.

87. Flotation of the stations due to groundwater uplift forces was critical for all non-CD stations. For CD stations because of the thicker walls and roof slabs flotation was not critical. Various solutions were provided in the non-CD stations; in Dhoby Ghaut station tension piles were provided under the box slab; in Toa Payoh, Novena, Tanjong Pagar and Outram Park stations counter-weight was added by way of cantilever toe slabs and/or shear binders; in Orchard station, in addition to tension piles, provision was made by way of a drainage gallery under the base slab to keep the water table low until the future development above is built.

Waterproofing

88. Waterproofing was provided only to the roofs of stations except for the three Engineer designed stations in phase IIA which are fully tanked. A stringent 5 $ml/m^2/h$ limit set on water leakage through exposed surfaces was achieved almost everywhere. Shrinkage cracks of more than 0·2 mm and all cracks that were leaking were pressure grouted irrespective of whether or not the allowable leakage criterion was met. Where exposed walls were within the public area, false walls with drains behind were provided in order to avoid damage to the architectural finishes by seepage.

89. In phase IIA underground stations, the designers specified an unbonded waterproofing system. It was found exceedingly difficult to achieve effective waterproofing with this system owing to its inherent weakness in that if a leak develops it is impossible to determine the point of entry of the water, and the lack of experience of local subcontractors with this type of system. In Marina Bay station, after considerable effort but with very slow progress, the contractor's proposal to change to a bonded membrane system was accepted.

CD design

90. For stations which were provided with CD facilities it was essential to ensure that the CD design did not in any way affect the operational capabilities of the system. CD design determined the thickness of the external walls which ranged between 1·35 and 2·0 m depending on the additional protection given by other elements such as diaphragm walls and anti-spalling plates, and the degree of protection required. Reinforcement requirements were mostly governed by peace-

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time loading rather than by CD loading except for the concourse slab, internal partition walls, ceiling hangers and other minor elements. However, the number of reinforcement layers, the shape of the reinforcement (CD main reinforcement has a hook at the end), lap lengths, shear links and other binders were governed by CD considerations which made the construction of CD stations more difficult (Fig. 18).

Viaduct

Structural form

91. Much effort was expended in careful assessment of the various factors that influence the detail of the cross-section in arriving at the structural form (Fig. 19). An important early decision was to locate the power cables on the inside between the tracks. This in turn determined the location of the third rail on the inside and the signalling and communication cabling on the outside because of the need to separate power and communication from interference considerations. It was decided to place the cables on brackets with a sun-shield rather than in troughing.

92. A walkway access along the viaduct was included to allow for daytime inspection of the track. At points and crossing areas, the central power cabling is diverted across the tracks to run under the walkways.

93. Design was carried out to BS5400 using RL standard loading. The 16 t maximum axle loading of the MRT stock (23 m long) is comparable with the 15·4 t maximum axle loading of the shorter vehicles used in the determination of RL standard loading. Conventional design analysis was used for the design of piled foundations, pile caps, circular piers, crossheads and box girders. Concrete grades were: precast prestressed box girders, 45–55; in situ spans prestresed, 45; cross-

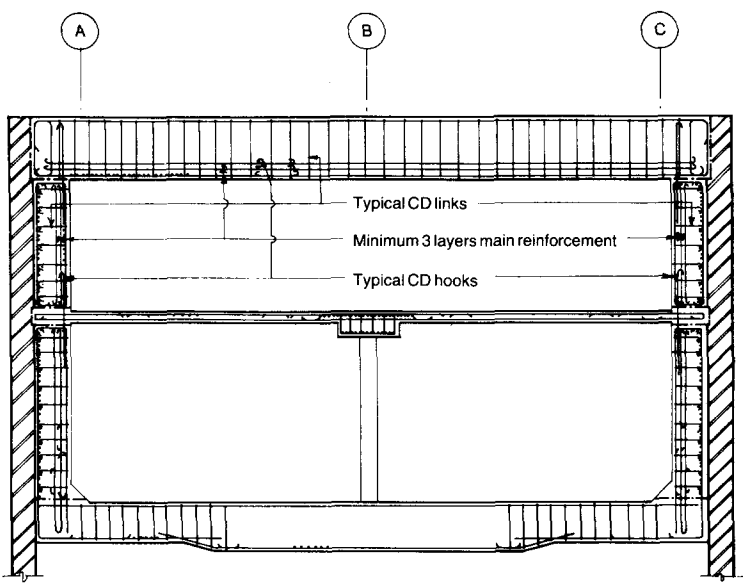


Fig. 18. CD reinforcement arrangment

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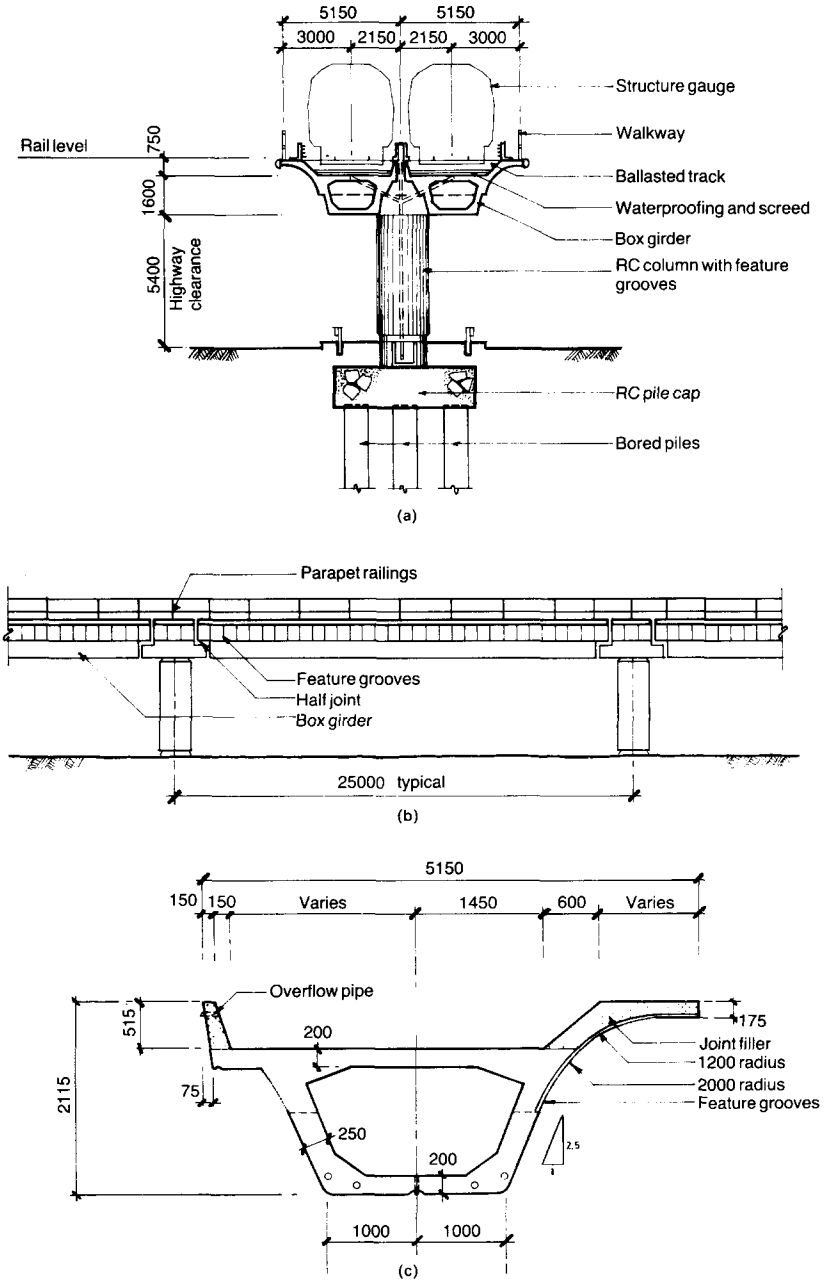


Fig. 19. Viaduct structure: (a) section; (b) elevation; (c) box girder section

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heads and columns, 40; piles and pile caps, 30–40. No structurally welded reinforcement was employed and no stress reversal under live load occurs. Hence fatigue was not of particular concern.

Bearings

94. For the simply supported spans of the viaduct, different concepts and types of bearings were adopted by the consultants (Fig. 20). On phase I, four laminated rubber bearings were used per beam with the horizontal loads resisted by shear pins; one end fixed and one end guided. On phase IA, four pot bearings were used per beam, fixed at one end and partially guided at the other. These bearings had additional top and bottom plates which helped in setting them in place and will permit easier removal and replacement in the future.

95. A different concept of bearings was adopted for phase IIB and copied, at the MRT Corporation's request, on phase IIA. Each beam was supported on three bearings only, one fixed pot bearing at one end and two partially guided pot bearings at the other. The design indicated that there would be significant transverse rotation of one crosshead in comparison with the adjacent crosshead under certain loading conditions. A different pattern of differential settlement of piles within one pile cap compared with the next may produce further differences in crosshead rotation. The viaduct beams are torsionally stiff and it was found with the four bearing arrangement that a large proportion of the load was being taken by three or sometimes only two bearings, depending on the load case. With the three bearing arrangement, the loads taken by each bearing are unaffected by any rotations of the crosshead, and compared with the four bearing arrangement, all three bearings could be designed for approximately the same load. So not only was there a saving in the number of bearings by 25%, the cost of bearings also reduced by approximately this amount.

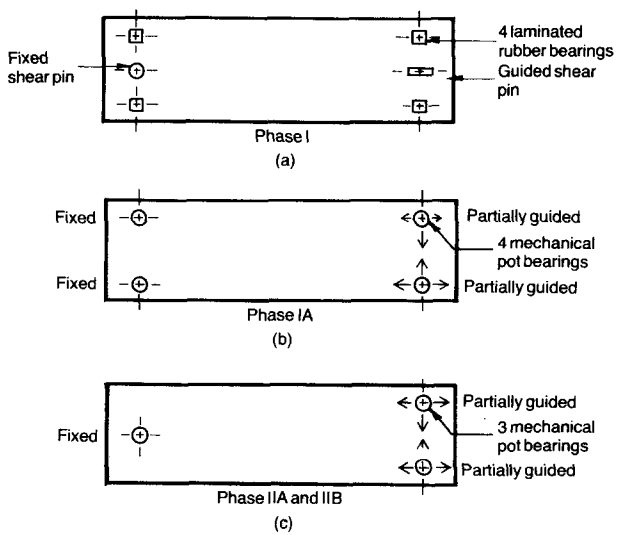


Fig. 20. Bearing arrangements

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96. BS5400 requires a derailment load of 30 kN/m to be considered on the very edge of the viaduct. If this were to be applied near the single bearing end, toppling of the beam would have occurred. To prevent this, stops were constructed on the crosshead ledge which arrested rotations of the beam at 1·3°. Another advantage of the three bearing system arose from the fact that the ratio of horizontal to vertical load was sufficiently low in all loading cases for the horizontal load to be resisted by friction alone. Two nominal shear bolts per bearing were provided to resist vibration and accidental impact as required by BS5400 but the fixing details were simplified and the criticality of tolerances on position became less important. With the three bearing arrangement, torsion in the beam increases but this was found to have little effect while savings could be achieved from reduced bending and torsion in the crosshead. The only disadvantage of the three bearing system is the inaccessibility of the fixed single bearing which will make replacement more difficult.

Load testing on viaduct beams

97. Full scale static and dynamic load tests were performed on typical viaduct beams to analyse and compare the actual behaviour of the beams against theoretical values.

98. In the static load test superimposed loads such as ballast, rails and parapets, were considered together with serviceability live loads specified in BS5400. In total a 130 kN/m uniformly distributed load and a 240 kN point load were applied. The loads were offset with respect to the centre line of the track to simulate centrifugal, nosing and lurching effects. Wind load was not considered. Surface mounted vibrating wire strain gauges were used to measure strain changes in the precast beam and in the crosshead. In addition, beam deflexion, column sway and relative movement between precast beam and the crosshead were also measured. Actual column sway and crosshead deflexion were found to be very close to the calculated values whereas beam deflexion, rotations and tensile and compressive stresses were within 16% of the expected values.

99. The dynamic test was carried out with an empty six car EMU. Each car had 4 axles and weighed 38 t, i.e. an axle load of 9·5 t compared with the design criterion of 16 t. Quarter bridge strain gauges of electrical resistance 120 Ω were attached to the beam surface for strain measurements. Deflexions were measured using strain gauge type displacement transducers. All the above instruments were connected to a high-speed data logging system which consists of signals, conditioners and high-speed scanning multiprogrammer. The data was recorded at a constant time interval of 0·005 s.

100. Before dynamic load testing, background signals without any loading were recorded, a static test with a stationary train and an ambient vibration test with the train departing at low speed were performed. Following this, dynamic load tests with the train moving at 30, 60 and 74 km/h respectively were performed. An emergency stop testing of the train moving at 70 km/h was also recorded.

101. The results showed that the strains and deflexions measured during the dynamic tests were smaller than the static load results. This phenomenon was attributed to the increased Young's moduli under dynamic conditions and to the high quality of the track which provided an especially smooth ride rather than a demonstration of worst case dynamic loads, e.g. with wheel flats.

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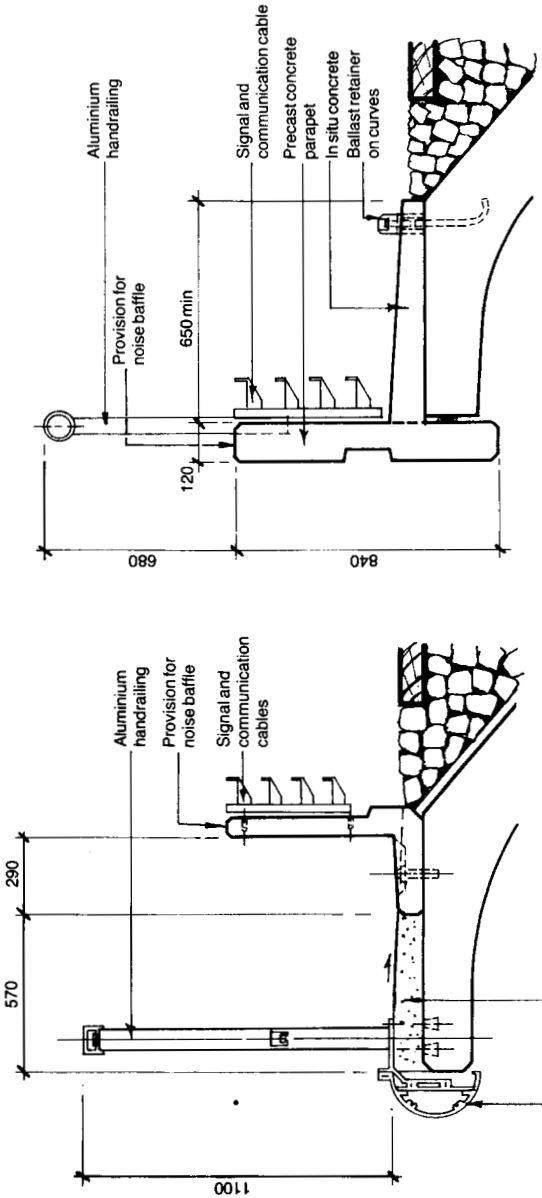


Fig. 21. Viaduct edge details

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Noise

102. There are clear indications that noise from the trains will not be a lasting nuisance to local residents. Experience on other modern railway systems shows that train noise is subjectively less intrusive than road noise once it becomes familiar. However, provision has been made for noise baffles to be fixed to the top of the wall holding the signalling cables (phase I, IA and IIB) or to the walkway (phase IIA). The presence of barriers would detract from the aesthetics of the viaduct and it is hoped that they will be avoided (Fig. 21).

Elevated stations

Structural framework

103. The system includes 26 elevated stations and one at-grade station. All elevated stations were included in the Engineer's design contracts. All stations have at least two levels—a ground level concourse area and an upper level platform. The structural elements are designed in reinforced concrete and prestressed concrete. However, structural frame arrangements are different. In phase I and IA stations, the station structure is completely separated from the viaduct structure (Fig. 22) whereas in phase IIA (Fig. 23) and IIB (Fig. 24), the platform main beams are supported off the viaduct columns. In these stations elastomeric bearings are provided below the platform beams for vibration isolation.

Roof types

104. Perhaps the most prominent feature that separates phase I, IA, IIA, IIB stations is their roof structure. Phase I and IA stations are conventional beam slab contruction. Phase IA has a lattice type extension to the roof overhanging the track to provide increased protection against rain (Fig. 25). Phase IIB stations display a local touch with Chinese and Malay type roofs constructed of structural steel frames (Figs 26 and 27). Phase IIA stations were designed with a reinforced concrete barrel arch roof of 23·6 m span (Fig. 28).

105. The barrel roof was analysed as a three-dimensional structure and the design is such that the longitudinal load transfer to columns at 25 m interval takes place by means of both the edge beams which carry 35% of the roof load and by the curved slab itself which carries the remaining 65%. This is made possible by the stiffness of the curved slab which is 3·5 m deep compared with the 1·4 m depth of the edge beam. Because of this design consideration the roof had to be cast in 25 m bays rather than 6·25 m.

Depots

106. The Singapore MRT has three depots. The main depot at Bishan (Fig. 29) is on a 35 ha plot and includes a 2 km long test track. Changi sub-depot which is located between Tanah Merah and Simei stations is on a 17 ha plot, and Ulu Pandan sub-depot which is basically only a stabling area for trains occupies 9 ha of land near Clementi on the west line. The facilities at each depot are summarized in Table 4.

107. The design problems that needed to be faced for each depot all concerned ground conditions and are described in the companion Paper 9430 on construction. At Bishan depot, the solution was to build a concrete deck on precast and

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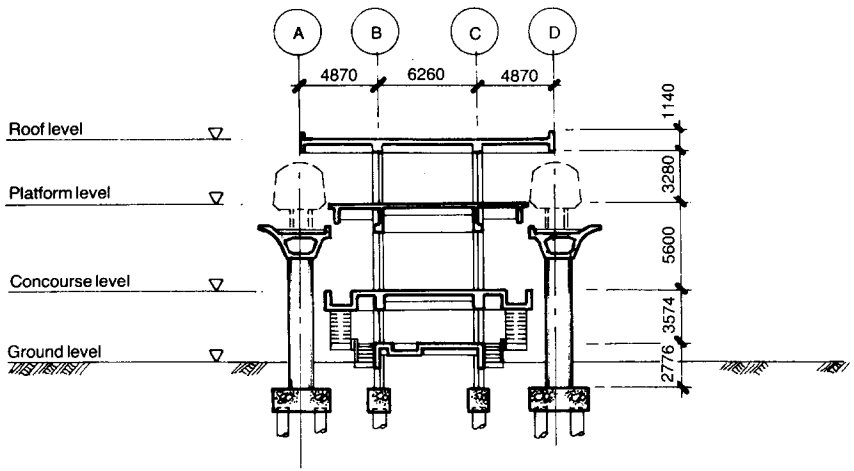


Fig. 22. Cross-section of phase I and IA station

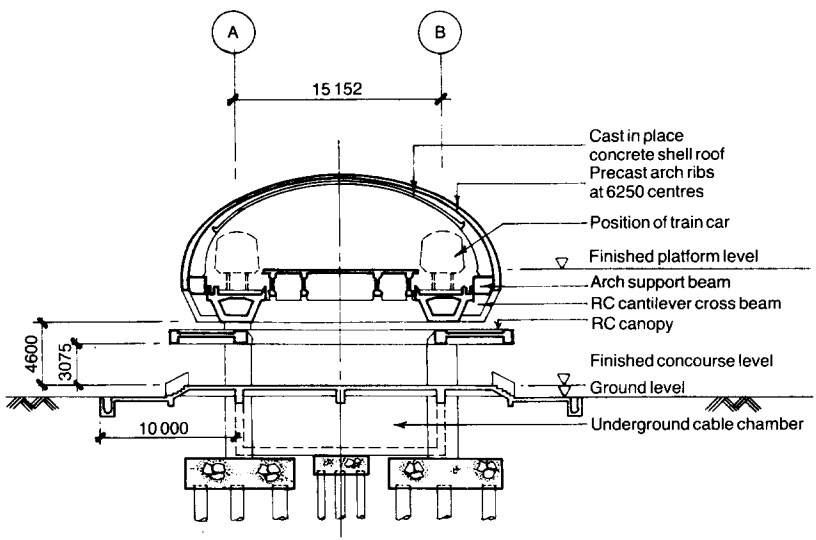


Fig. 23. Cross-section of phase IIA section

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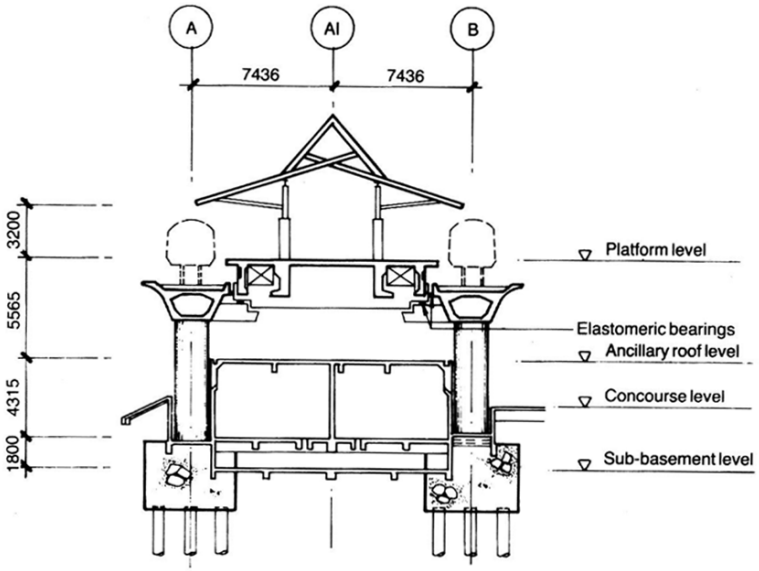


Fig. 24. Cross-section of phase 11B station



Fig. 25. Clementi station

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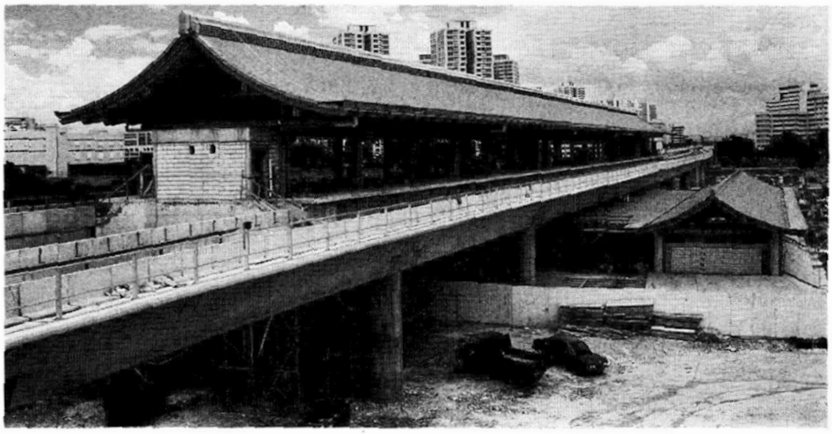


Fig. 26. Chinese Garden station



Fig. 27. Eunos station

steel piles as other methods to overcome the long-term settlement of the peaty clay were determined to be unsuitable or unable to be achieved in the time-scale available.

108. The concrete deck design involved a combination of precast and in situ concrete. An 8 m × 4·5 m grid was selected and generally concrete piles were driven in pairs at each grid point. Small pile caps were formed and precast beams resting on plain neoprene bearing pads span the 8 m between caps. Precast planks 100 mm thick with reinforcement stirrups protruding span between the beams and are covered with 200 mm of in situ concrete (Fig. 30).

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Fig. 28. Tampines station

109. There are three main sewers crossing the site and here steel H-piles were used instead of precast concrete in order to minimize ground movement. Three drainage canals had to be spanned which involved the use of 18 m and 22 m long prestressed I-beams on continuous strip abutments. Altogether approximately 15 000 piles, 3500 precast standard beams, 18 000 planks and 550 prestressed I-beams were needed to provide 120 000 m^2 of deck.

Other structures

110. Six subways that connect the underground stations to adjacent commercial buildings or bus depots have been constructed and others have been allowed for in the design of the underground stations. Future subways are expected to be constructed concurrently with future development planned around the stations. Other structures include a three storey 56 m × 46 m operation control centre and four 66 kV intake substations. These buildings are all conventional reinforced concrete structures.

Stray current protection

111. A phenomenon of modern railways which use the running rail as the return path for the electric current supplied to the trains is stray currents. As the rail is not thoroughly insulated but relies on a thin plastic pad to give some electrical isolation in the tunnel section and the insulation properties of timber sleepers in the elevated and at-grade sections, current leakage from the rails into the structure and into the ground is inevitable. The problem has been minimized by adopting a heavy 60 kg/m rail and by the maintenance department carrying out regular cleaning operations in the tunnel sections to avoid the build up of dirt, rust and dampness around the rail fastenings. Typically, an earth to rail resistance of 50 Ωkm in the tunnel and 100 Ωkm on the viaduct on a dry day has been achieved.

112. Stray currents cause damage to reinforcement by electrolytic corrosion and two methods have been adopted in the design to minimize their effect. The

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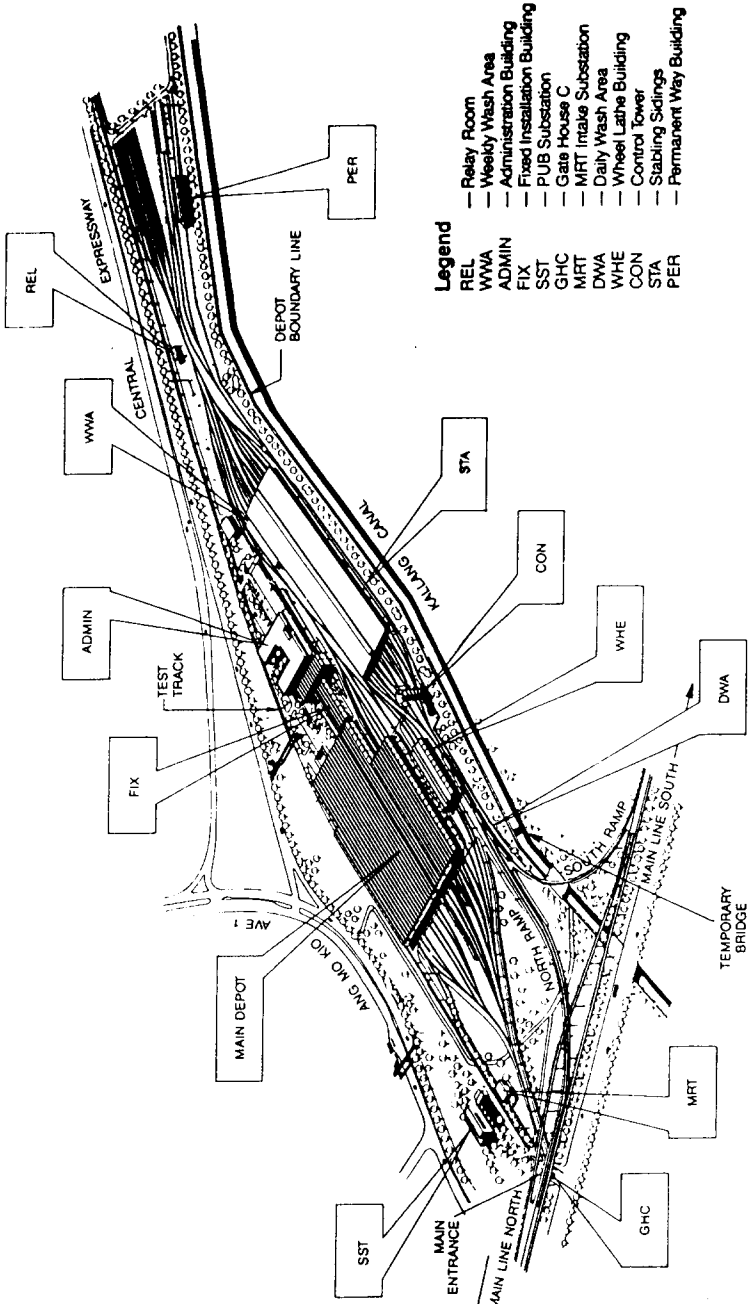


Fig. 29. Bishan depot general view

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Table 4. Depot facilities

Facilities	Depots		
	Bishan	Ulu Pandan	Changi
Main depot facilities			
Routine inspection	Yes	Yes	Yes
Car servicing	Yes	Yes	Yes
Quick repair	Yes	No	Yes
Bogie exchange	Yes	No	Yes
Overhaul and repair of main components	Yes	No	No
Test bays	Yes	No	No
Stores	Yes	No	Small stores
Secondary facilities			
Daily wash plants	Yes	Yes	Yes
Weekly wash plants	Yes	No	No
Under-floor wheel lathe	Yes	No	Planned
Permanent way and signalling maintenance facilities	Yes Yes	Only staff accommodation and small store	Yes
Internal washing	Yes	Yes	Yes
Diesel fuel filling stations	Yes	Temporary	Yes
Fixed installation	Yes	Examination only	No
Training facilities	Yes	No	No
Cash handling	Yes	No	No
Test track	Yes	No	No
Train shelters	Yes	Yes	Yes
Administration	Yes	Small offices	Yes

methods rely on different principles: one on reduction of the amount of stray current that any piece of reinforcement is exposed to; the second on providing a return circuit within the reinforcement of the structure. The two methods result in very different designs.

113. For the former, the structure is divided into small units, each unit being electrically separated from the next. Because each unit is small, the amount of stray current attracted to each unit will be small and its effect sufficiently minute not to cause concern. This principle is ideally suited to the bored tunnel segments and these were detailed such that where steel bolt pockets were used, the steel of the pocket and its ties were separated by means of plastic spacers from the reinforcement cage.

114. Elsewhere the second method was adopted. In cut-and-cover tunnels, underground stations and the like, this involved making all reinforcement within 3·25 m of the track centre line continuous by welding. On the viaducts it involved the placing of a steel reinforcing fabric either on the top face of the box beams (phase I, IA and IIA) or within the screed that protects the waterproofing (phase IIB and IIC). Copper jumper cables were needed at joints between beams and crosshead, thermit welded to the mesh. Reinforcement in the track-bed concrete was also made continuous by welding with jumper cables placed at floating slab track areas where there was no track-bed reinforcement. The continuous mesh will

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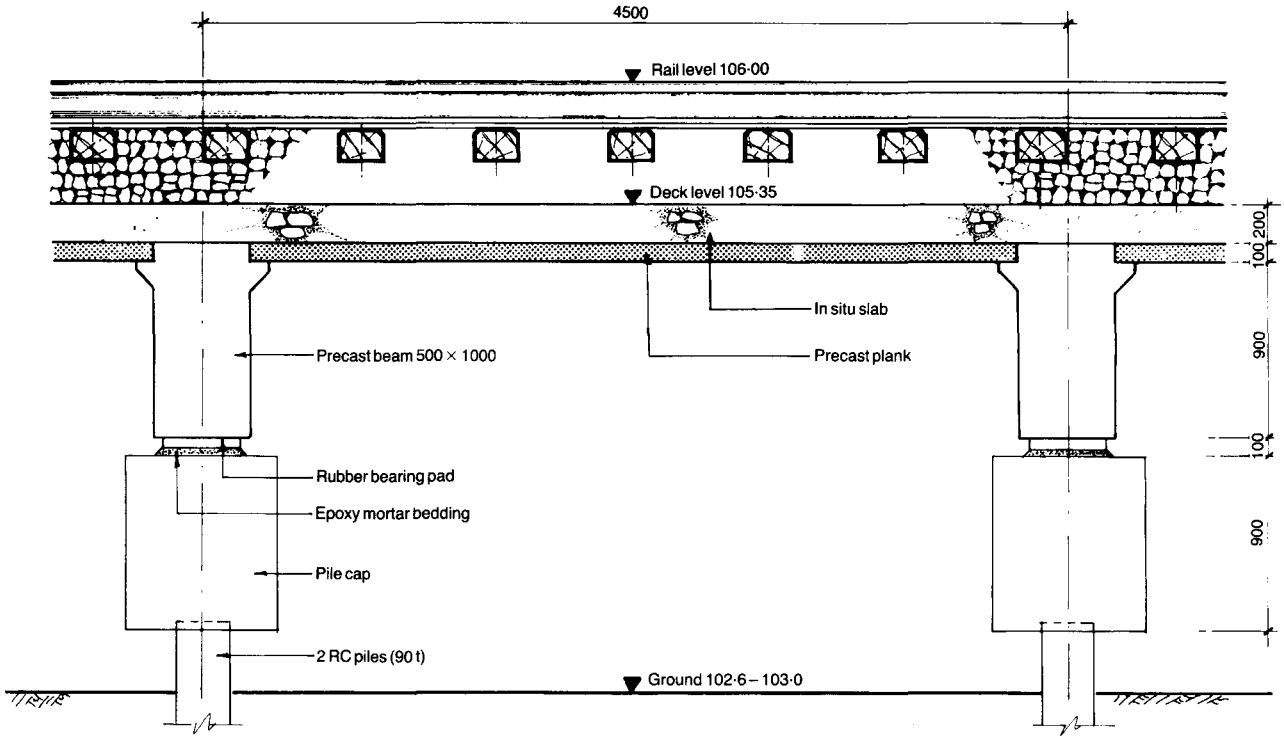


Fig. 30. Bishan depot deck system

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act as a sieve to allow the drainge of a significant portion of the stray currents. Terminals connected to the reinforcement are provided at the traction substations to allow the drainage circuit to be closed and to allow monitoring of current flow.

115. Voltage monitoring will be carried out at specific locations along the railway. Earth potential reference electrodes have been installed in the tunnels whereas on the viaduct the handrail has been made continuous and connected to earth for lightning protection purposes and will be used as an earth potential. Monitoring points are typically at 200 m centres.

Lessons learnt

116. The decision to leave the individual design-and-construct contractors to draw up their own materials and worksmanship specification was made to avoid inhibiting designers recruited worldwide. In practice contractors having been selected by international tender were intent on providing minimum standards subject only to compliance with the MRT Corporation's performance specifications. To obtain a workmanlike structure with a guaranteed life of 75/100 years under these conditions imposes a significant load on the Engineer and his staff.

117. In the Engineer's design contracts the limited design period tended to force consultants to adopt over-conservative concepts. Fixed fee contracts afforded little incentive to produce optimum design solutions. It is therefore essential to plan forward effectively to allow a reasonable minimum (circa one year) design period and to negotiate a realistic fee.

118. The need for equipment design to precede civil design was emphasized by the successive modifications requested by E&M contractors throughout the civil design and construction period. The problem was exacerbated by the decision to entrust design and site co-ordination to the civil design–construct contractor. He was obliged to produce combined service drawings and drawings detailing openings and service penetrations in the structures from information gathered from electrical and mechanical system-wide contractors but with little effective control on the validity of the data. It is preferable for the Employer to undertake and strictly control all design co-ordination.

119. From the outset this £1400 million project was based on a fast-track programme and the acceleration of phase II by two years increased the pressure on the MRT Corporation's staff. Despite this no delays were experienced in the construction due to the design process. The final judgement on the system will be pronounced by the people of Singapore when it is fully operational. Present passenger response indicates enthusiastic acceptance.

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